

**OBJECT STORAGE
USER GUIDE
OF
GOVERNMENT CLOUD
INFRASTRUCTURE SERVICES
(GCIS)
FOR
THE OFFICE OF THE GOVERNMENT CHIEF
INFORMATION OFFICER**

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DEC 2021

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Amendment History					
Change Number	Revision Description	Pages Affected on Respective Version	Revision / Version Number	Date	Approval Reference
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1.1	Add steps for adding root certificate to trusted list in Section 6.3			03-Sep-2021	
1.2	Add information about the production environment in Section 2, 3.and 5			21-Dec-2021	

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1. PURPOSE

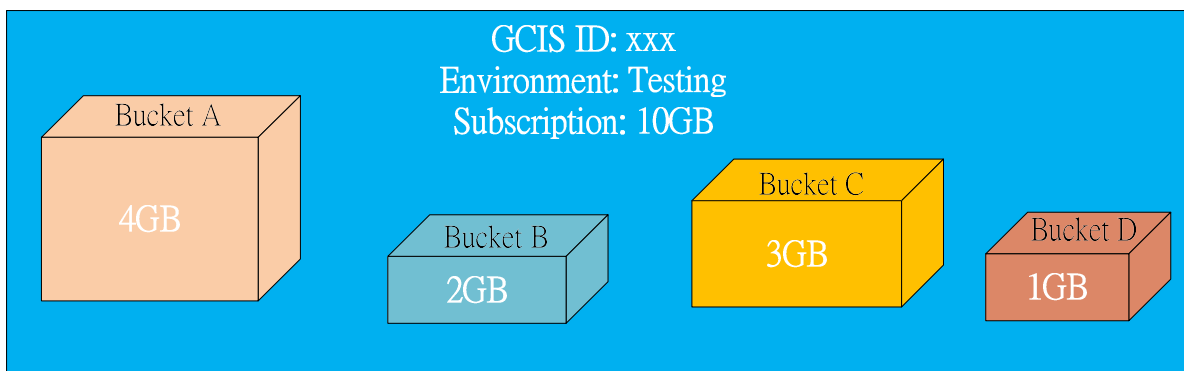
This user guide demonstrates the basic usage and operation of GCIS object storage. Hitachi Content Platform (HCP) SAN-attached configuration is deployed in GCIS object storage. After subscribing GCIS object storage, B/Ds can use the tools provided by HCP including Tenant Management Console, HCP Namespace Browser for the self-service administration of the object storage. This guide will show how to create buckets, store and retrieve objects, and manage permissions on resources.

2. GCIS OBJECT STORAGE OVERVIEW

GCIS offer object-based storage to B/Ds in order to store, archive or backup high volume of static or unstructured data. B/Ds can also use the object storage as a shared data or file storage for their application access. For data protection, all object data are stored in the SAN storage system which is configured as RAID-6 and they are also protected by two redundant copies in two independent sites. To cater different performance requirements, auto-tiering disk storage is deployed in the SAN storage system. In addition, all object storage data are encrypted by 256 bit AES encryption where the master key is stored in external Key Management Service (KMS).

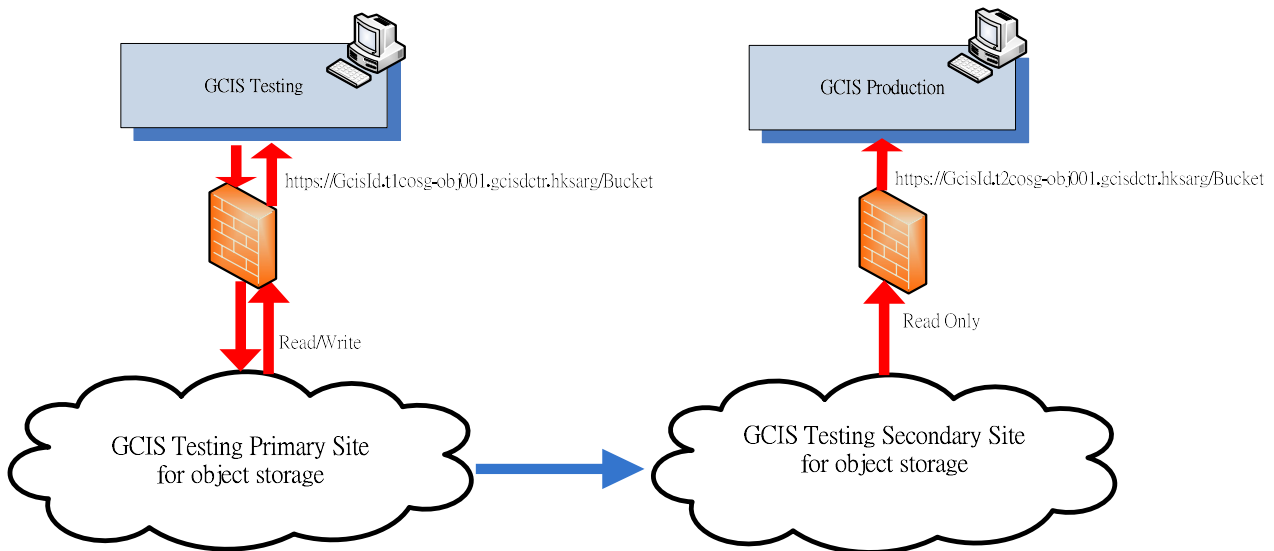
GCIS object storage is Amazon S3 compatible so that B/Ds can access the object data via Amazon S3 APIs for most of the access operations. For the list of the supported Amazon S3 compatible APIs, please refer to Section 14 for details. In addition, B/Ds can use S3 compatible third party tools such as S3 browser to access the data as well. In Section 13, there is a guideline to show how to access GCIS object storage using S3 browser.

GCIS tenant can subscribe GCIS object storage with a desired storage space (quota) per environment, ie. testing or production and the subscribed storage is therefore the usable space for storing object data. Within the subscription storage space, B/Ds can allow creating at most 20 buckets. B/Ds must assign a hard quota to the bucket when it is created and the bucket quota will be derived from the total subscribed quota. For example, a tenant subscribes 10GB object storage in testing environment and four buckets with total size 10GB have been created as illustrated in the following diagram. Then, the tenant cannot create more buckets in the environment but the tenant can adjust the bucket quota dynamically if the actual usage is not reached the defined bucket quota in order to free up the quota.

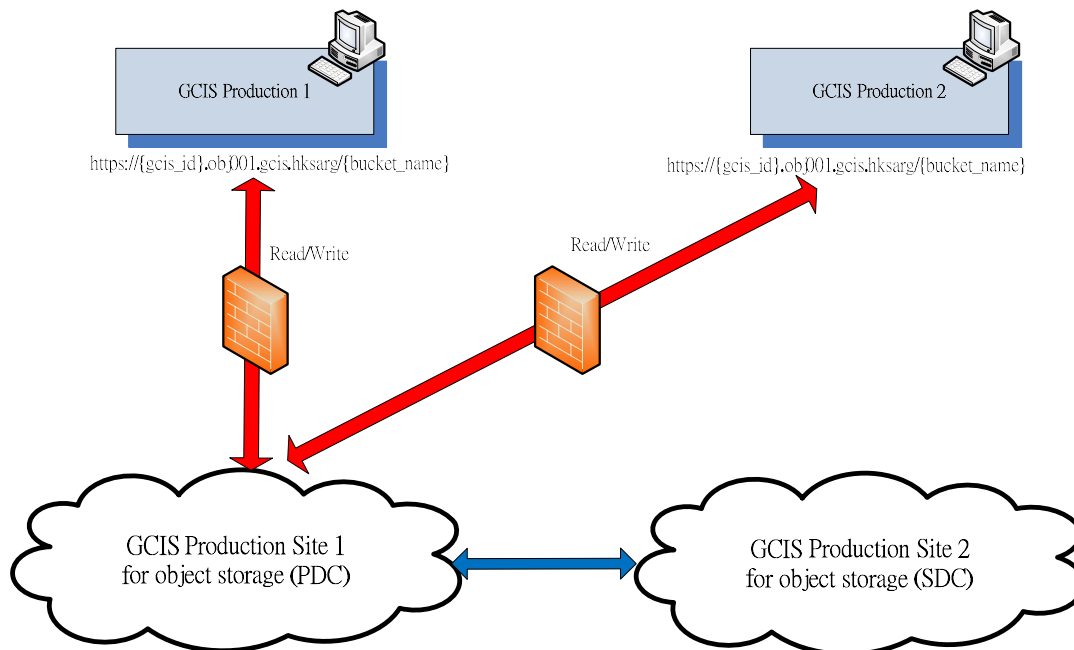


GCIS object storage is a backup free storage and traditional backup is not offered. Data replication & versioning feature is used to serve the backup purpose instead. In all GCIS environment, all object data are replicated to the remote site in asynchronous mode and there is around 5 to 10 minutes delay in the remote site after writing the object data normally.

In testing environment, Active-Passive is configured for the data replication. B/Ds can read or write object data in the primary site and only read operation is allowed in the secondary site. For object data in the secondary site, B/Ds can read the data from GCIS production if firewall request is released from B/Ds' production servers to the object storage in the secondary site. The arrangement is aimed to facilitate data transfer from testing to production environment, especially used in program deployment. The following diagram illustrates the object storage replication flow in the testing environment.



In production environment, hot standby mode is designed for the object storage service. All object storage data traffic under a tenant will be redirected to the primary data centre (PDC). In case the PDC is failed, all data traffic will be redirected to the secondary data centre (SDC) automatically. All writing of object data in any production site will be replicated to the remote site for the disaster recovery purpose. Although the replication is asynchronous, B/Ds' applications can always retrieve the up-to-date object data in the hot standby mode design.



3. POINTS TO NOTE

1. The term “Namespace” in Hitachi Content Platform is equivalent to “Bucket” in Amazon Simple Storage (S3).
2. B/Ds must ensure the uploaded object files have been scanned through anti-malware tools prior to storing in GCIS object storage.
3. An administrative account and password are created for each tenant. For resetting the password,

- please contact GCIS Client Advisory (CA) in office hour and GCIS OCC in non-office hour.
- For all user accounts other than the tenant administrative account, they can be created and managed by tenant itself through the HCP management console.
 - Replication is enabled by default for improving data availability. Please do not alert any setting of replication to ensure your data is replicated properly.
 - The production failover is controlled by Global Traffic Manager (GTM) that will update the DNS record automatically in case of site failure. B/Ds should ensure that the application should not set forever or too long DNS caching timeout in order to refresh the updated IP address.
 - Please read GCIS object storage house rules for other details.

4. ABBREVIATION AND ACRONYM

Abbreviation	Description
B/Ds	Bureaux and Departments
GCIS	Government Cloud Infrastructure Services
HCP	Hitachi Content Platform
Amazon S3	Amazon Simple Storage Service
REST	Representational State Transfer
API	Application Programming Interface
GUI	Graphical User Interface
ACL	Access Control List
GCIS SSP	GCIS Self-Service Portal

5. BASIC INFORMATION FOR GCIS OBJECT STORAGE CONNECTION

5.1 ADMINISTRATIVE OBJECT STORAGE USER ACCOUNT

After GCIS object storage Provisioning, an object storage administrative user account in form of {gcis_id}_admin will be created. Tenant can login the HCP Tenant Management Console to create bucket, manage bucket quota, create user account and manage bucket ACL, etc.

Object Storage Admin User Name	{gcis_id}_admin
Object Storage Admin User Password	{provided after GCIS object storage provisioning}
HCP Tenant Management Console	https://{gcis_id}.tlcosg-obj001.gcisdctr.hksarg:8000 (Testing Environment) https://{gcis_id}.obj001.gcis.hksarg:8000 (Production Environment)

5.2 FIREWALL REQUEST

Tenant should submit Other Change request to GCIS to register their GCIS SSL VPN account(s) in order to access the Object Storage Tenant Management Console.

Sample firewall configuration request:

VPN User Name	Destination Service/Port	Objective
To be filled by Tenant	TCP/443,8000	Testing / Production Object Storage Tenant Management Console Access

For application access to GCIS object storage, they should provide source IP information and submit firewall request through Other Changes in GCIS SSP.

Sample firewall configuration request for IaaS client:

Source		Destination Service/ Port	Objective / Purpose / Project Code
VM Name (fill in IP if it is service IP)	IP Address		
To be filled by Tenant	To be filled by Tenant	TCP/443	Testing / Production Object Storage Access

Sample firewall configuration request for CaaS client:

Source	Destination Service/ Port	Objective / Purpose / Project Code
Tenant Egress IP Address		
To be filled by Tenant	TCP/443	Testing / Production Object Storage Access

6. ACCESS TO HCP TENANT MANAGEMENT CONSOLE

6.1 PRE-REQUISITE

Tenant should submit firewall request to GCIS to register GCIS SSL VPN account(s) in order to grant the network connection to the HCP Tenant Management Console.

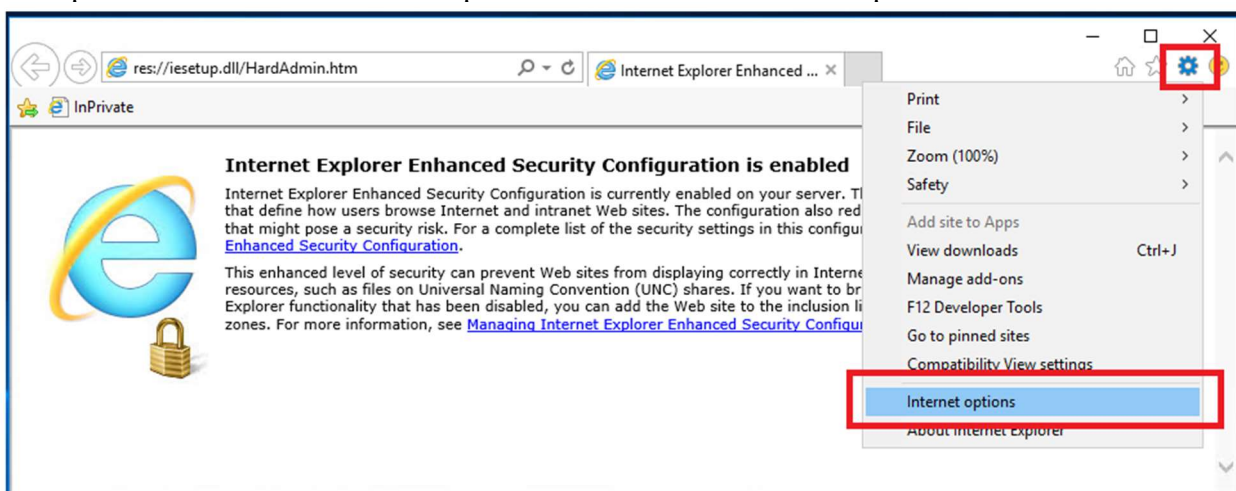
6.2 LOGIN SSLVPN

Use FortiClient login to GCIS testing environment VPN. Please consult Network team about how to login SSL VPN.

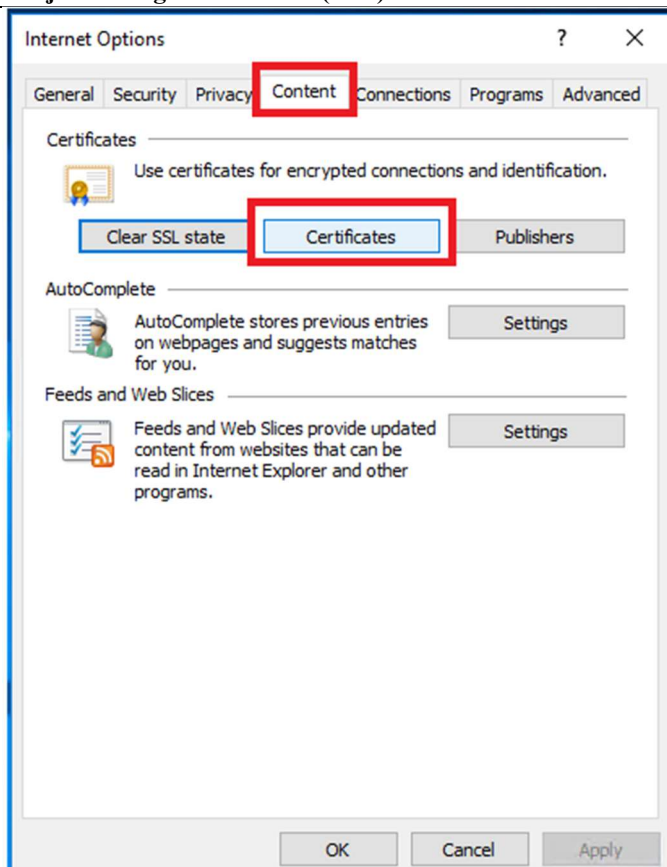
6.3 ADD GCIS ROOT CA CERTIFICATE TO TRUSTED LIST

In order to avoid security warning message appears when accessing HCP Tenant Management Console using browser, you should add GCIS root CA certificate and intermediate certificate to the trusted list. GCIS root CA certificate file in the format .der will be provided after subscription. The following shows the steps of how you can add the root CA certificate to the trusted list in Windows environment.

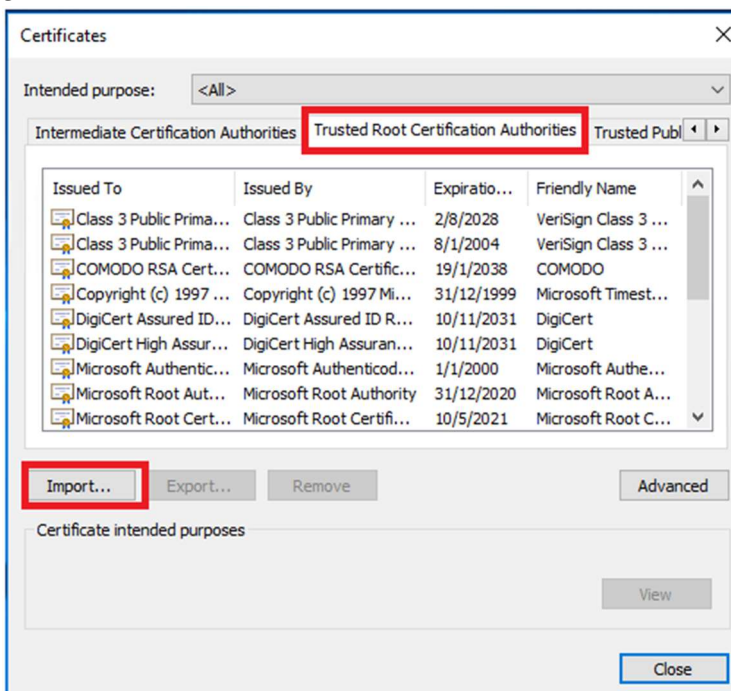
1. Open IE browser. Click on the option icon and select “Internet options”.



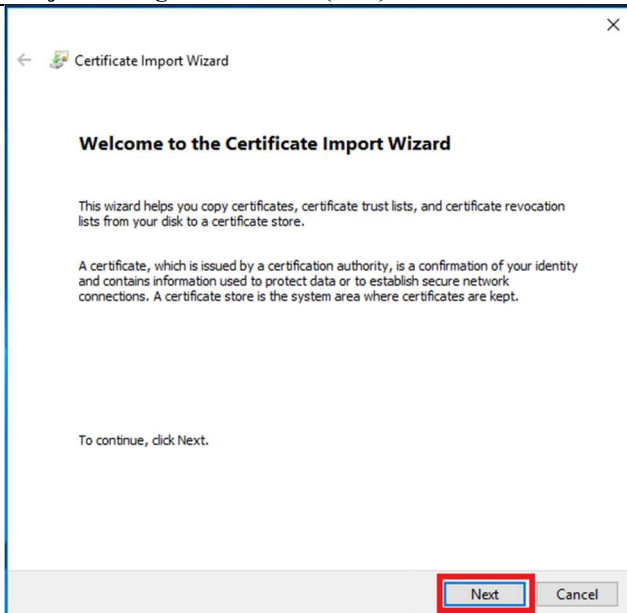
2. Select “Content” tab and click “Certificates” button.



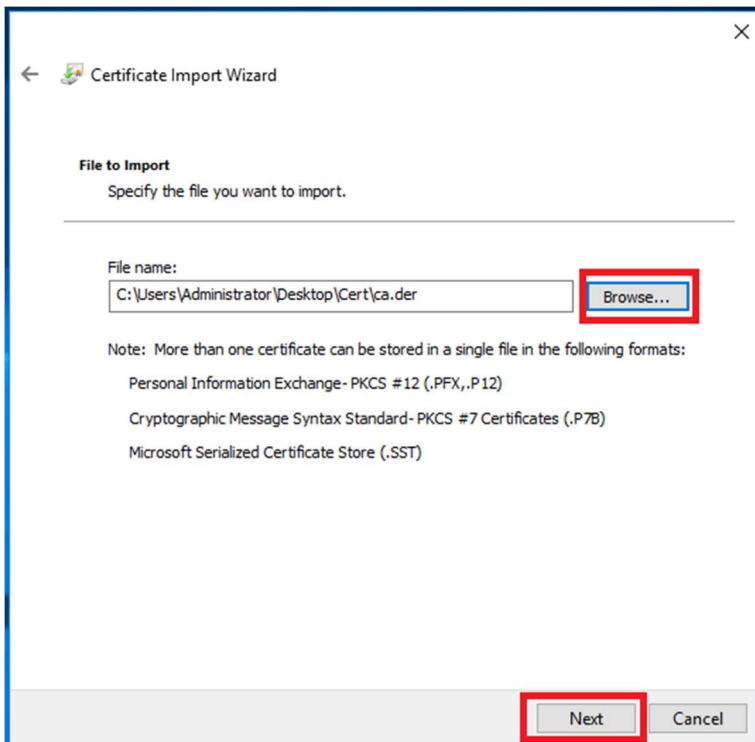
3. Select “Trusted Root Certification Authorities” tab and click “Import...” button.



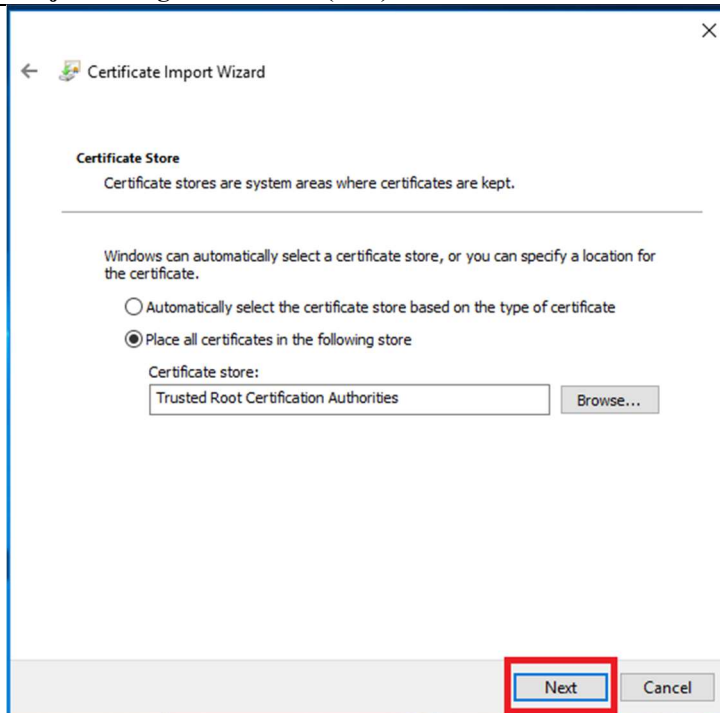
4. Click “Next” button to continue.



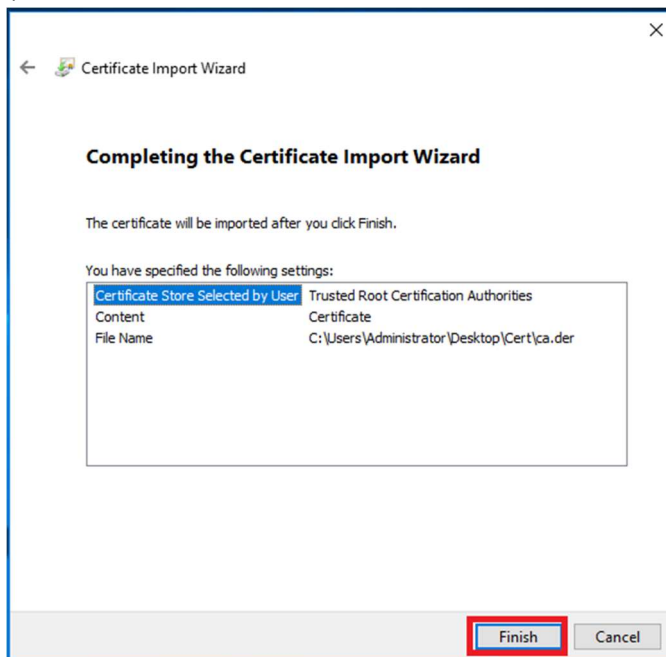
5. Click “Browse..” button and select the GCIS root CA certificate. Click “Next” button to go on.



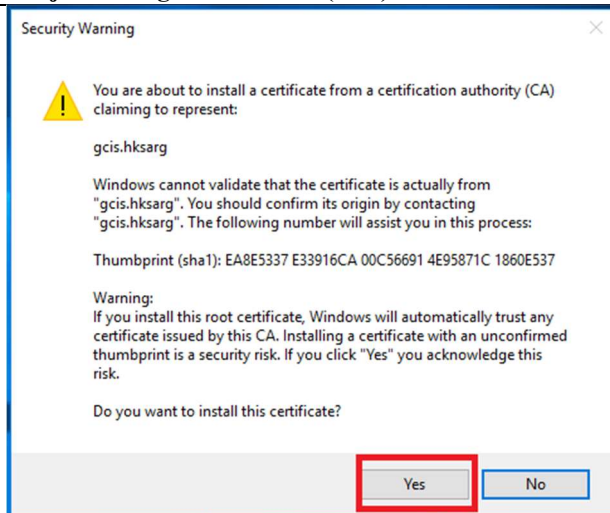
6. Click “Next” button to go to next step.



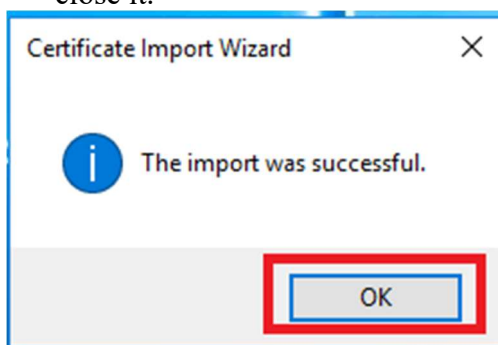
7. Click “Finish” button.



8. A security warning message will be shown. Click “Yes” button to continue.



9. The root certificate is added to the trusted list and a successful message will be show. Click “OK” to close it.



10. Please repeat from step 3 by selecting “Intermediate Certification Authorities” to import the intermediate certificate.

6.4 ACCESS HCP TENANT MANAGEMENT CONSOLE

Open your browser and enter HCP Tenant Management Console URL. Enter the administrative user name and password provided by GCIS into “Username” and “Password” field respectively and Click “Log In” to proceed.

6.5 CHANGE PASSWORD FOR THE FIRST SUCCESS LOGIN

You are required to change the password for the first success login. Please enter your old password in “Existing Password” field, and new password in “New Password” and “Confirm New Password” fields.

Click “Update Password” to change your password.

The screenshot shows the GCIS Object Storage Tenant Management Console interface. The top navigation bar includes 'Security' and 'Monitoring' tabs. The user is logged in as 'gid01_admin'. A message states: 'You are required to change your password at this time.' Below this, the title is 'Change Password for User: gid01_admin'. The form contains three input fields: 'Existing Password', 'New Password', and 'Confirm New Password'. At the bottom right, there are two buttons: 'Update Password' (orange) and 'Cancel' (gray).

After successful update of your password, a new authorization token is generated and shown on the page. The authorization token is used for accessing Object Storage through Amazon S3 API. Please record the authorization token for later use.

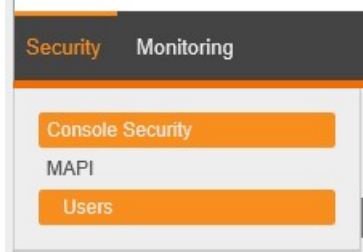
This screenshot shows the same 'Change Password' form as the previous one, but with a success message displayed at the top in a green box: 'Successfully updated user account. New authorization token: Z2lkMDFyW4==:161ebd7d45089b3446ee4e0d86dbcf92'. A 'Dismiss' button is next to the message. The form fields and buttons remain the same.

7. CREATE A BUCKET

Every object in GCIS object storage is stored in a bucket. Therefore you must create a bucket before you can store data to GCIS object storage. In order to create a bucket, you must be granted the “Administrator” role. You also need to create a user with permissions to work on the bucket.

7.1 ADD ADMINISTRATOR ROLE TO A USER

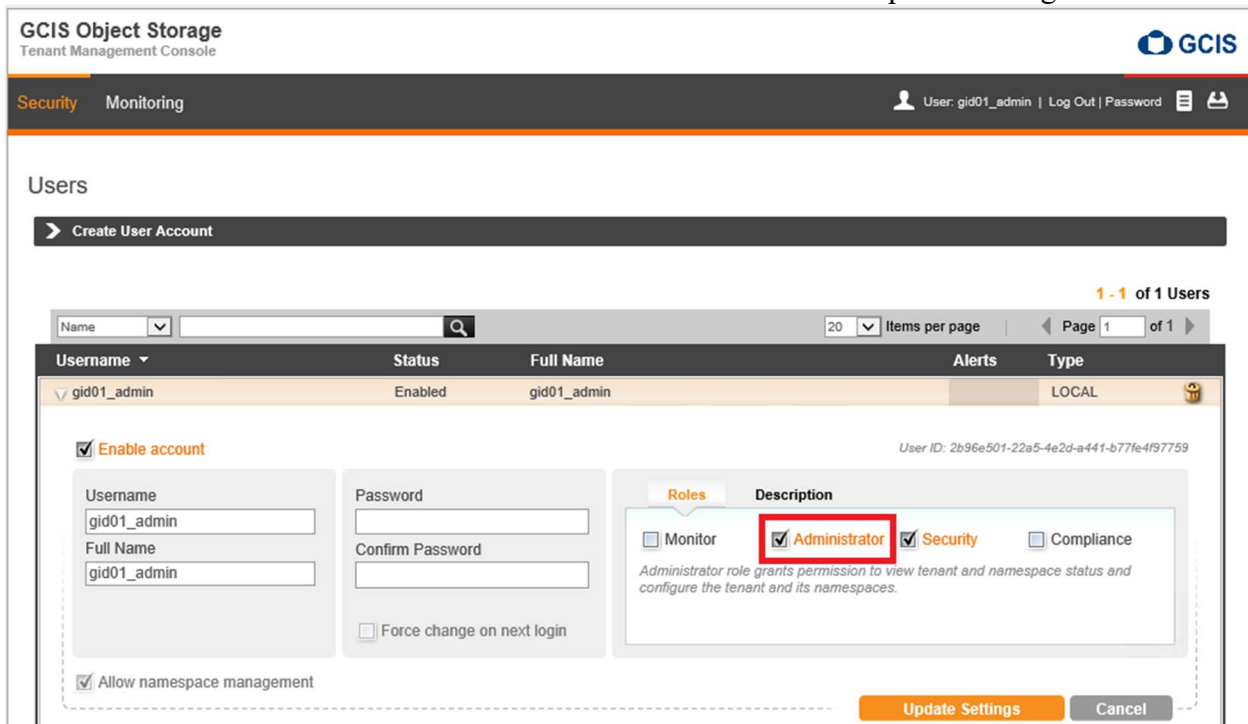
1. Log on to HCP Tenant Management Console. Click “Security” on the top menu bar. Choose “Users”



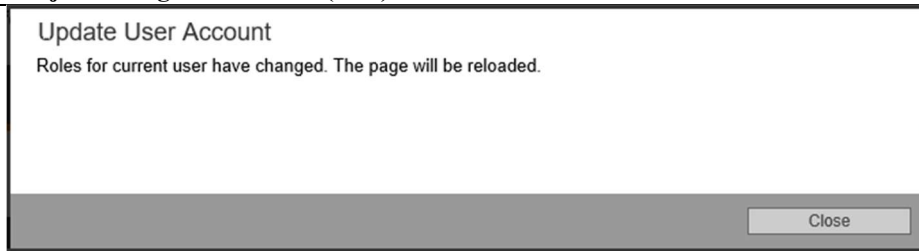
2. Choose the user name you want to add the role. You can first add the role to the administrative account first.



3. Click the check box beside the “Administrator” role and click “Update Settings” to add the role.



4. A pop up will be shown for the update success. Click “close” to continue.

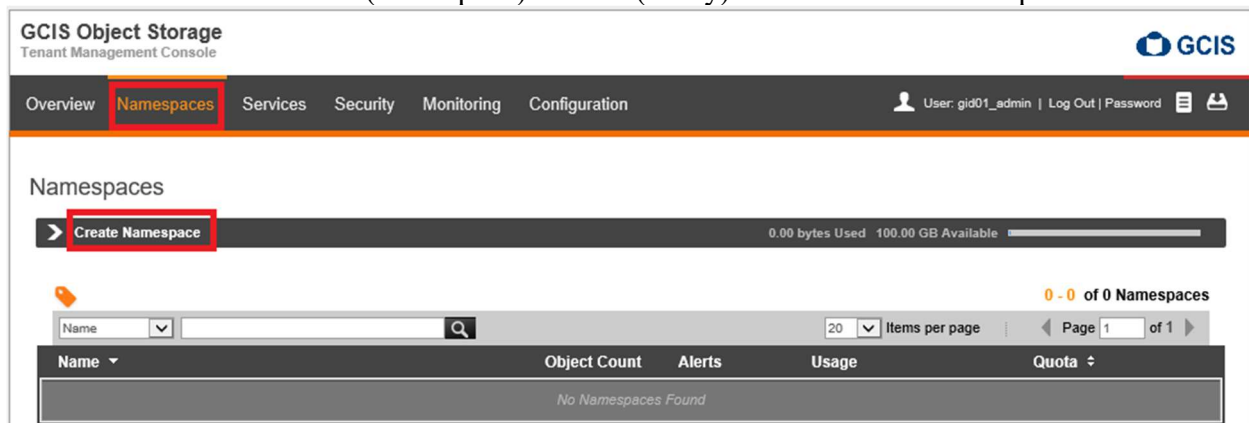


7.2 CREATE A BUCKET

Next you can create a bucket for storing data. In HCP, a bucket is called namespace. Tenant can create the bucket by using HCP Tenant Management Console.

If API is used to create the bucket, either S3 API call or HCP Management API can be used. Please note that the bucket will be assigned with a default quota (50GB) if S3 API is used. The default quota setting and existing bucket quota can be changed in the HCP Tenant Management Console. To create the bucket with specific feature provided by HCP, tenant should use HCP management API. For the API details, please refer to the HCP Management API Reference in the public web site.

1. From the HCP Tenant Management Console, click “Namespaces” on the top menu bar. This page shows the list of bucket (namespace) created (if any). Click “Create Namespace” to create a bucket.



2. Enter bucket name in “Namespace Name” field. In GCIS object storage, bucket quota is referred to hard quota; enter desired bucket quota in “Hard Quota”.

You may want to enable the bucket to keep multiple versions of the objects. Check “On” in the “Versioning” to enable versioning feature.

When the versioning is turned on, you should set the Version Pruning in terms of days. All old versions of objects will be deleted automatically after the pruning period. Please note that the housekeeping job runs at 0:00 every day.

If you disable Version Pruning, all object versions, including deleted versions, will remain in the bucket until you enable pruning or the objects are purged.

For details of versioning and pruning setting, please refer to Section 10.2.

Create Namespace 92.00 KB Used 100.00 GB Available

Namespace Name: (2 namespaces used 18 namespaces available)

Namespace Owner (Optional):

Local ☒ Active Directory ☐

Description (Optional):

Hash Algorithm: SHA-256

Hard Quota: 50.00 GB (Allocated: 2.00 GB available: 98.00 GB)

Soft Quota: 85 %

Tags: Optional

Configure features for this namespace

Cloud Optimized: ☐ Off ☒ On

Retention Mode: ☒ Enterprise ☐ Compliance

Search: ☒ Off ☐ On

Versioning: ☐ Off ☒ On

Version Pruning: ☒ Prune versions older than 30 days

Turning on cloud optimization prevents the CIFS, NFS, WebDAV, and SMTP protocols from being enabled for the namespace. While the namespace contains any objects, cloud optimization cannot be turned off.

Create Namespace Cancel

3. A new bucket is created and it is added to the list of the namespace.

GCIS Object Storage Tenant Management Console

Overview Namespaces Services Security Monitoring Configuration

User: gld01_admin | Log Out | Password

Successfully created namespace doc. Dismiss

Namespaces

Create Namespace 0.00 bytes Used 100.00 GB Available

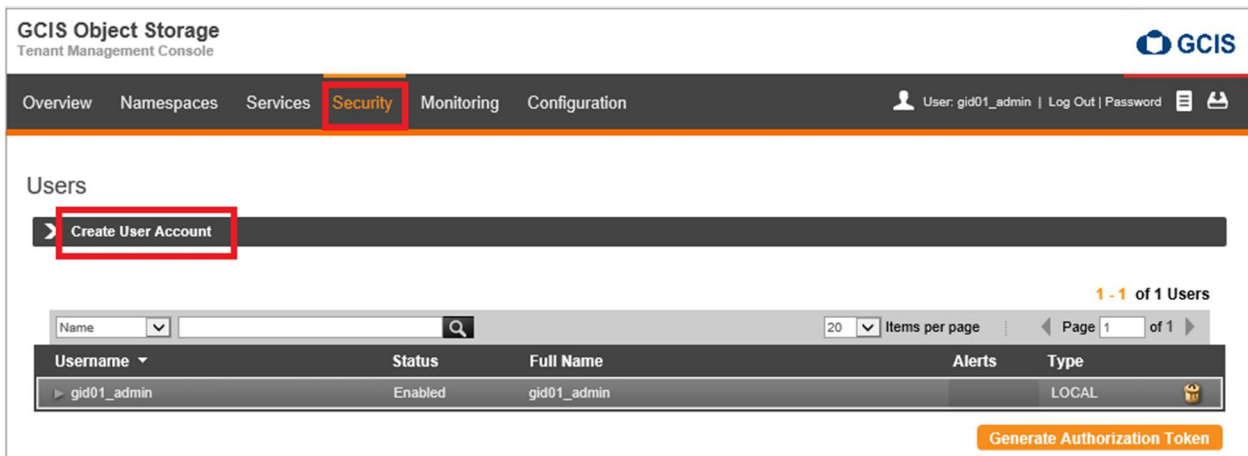
1 - 1 of 1 Namespaces

Name	Object Count	Alerts	Usage	Quota
doc	0		0.00 bytes	of 50.00 GB

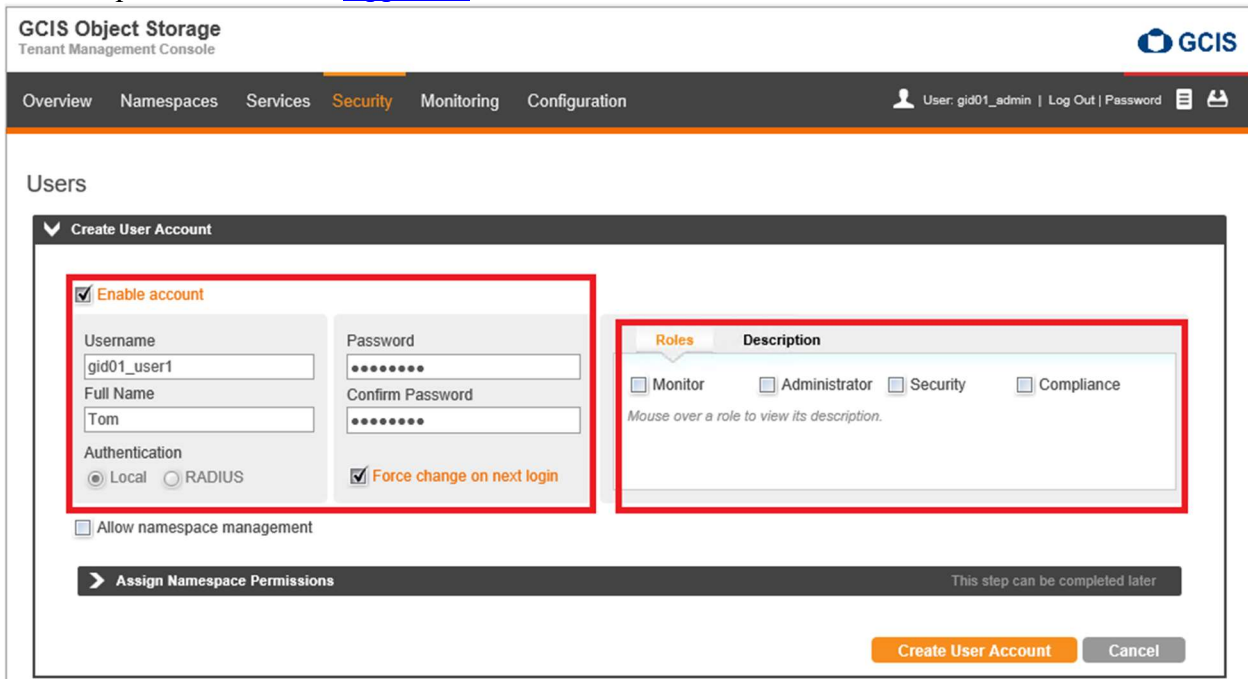
8. CREATE USER TO WORK WITH BUCKET

Next you should create a user account to work on the bucket. The user should have enough permission to access the bucket.

1. In HCP Tenant Management Console, click “Security” on the top menu bar and then click “Create User Account”.



2. Enter the “Username”, “Full Name”, “Password” and “Confirm Password”. Also click on “Force change on next login” to force user to change password at the first success log on for the sake of security. Leave all checkboxes in “Roles”. In option, you can assign additional roles to the user if it is required. Please see [Appendix](#) for the details of the roles



3. Next you have to assign permission to user to access the bucket. Click “Assign Namespace Permission”. Click on the bucket name (E.g. “doc”). Assign the desired data access permissions to user by clicking on the corresponding check box.

In general, for normal application usage, the system role of Monitor, Administrator, Security and Compliance is not required and B/Ds only assign the relevant bucket permission to the user. For example, you may create a user having “Browse”, “Read”, “Write”, “Delete” and “Search” permission for general application use. On the other hand, you can follow your business requirement to create different users for different roles. For example, create a read-write user to have “Read”, “Write” and “Delete” permission, create a read-only user to have “Browse”, “Read” and “Search” permission. For further details on the permission, please refer to Appendix.

After you have finished all settings, click “Create User Account” to create the user.

Users

Create User Account

☒ Enable account

Username:

Full Name:

Authentication: ☒ Local ☐ RADIUS

Force change on next login: ☒

Allow namespace management: ☐

Assign Namespace Permissions This step can be completed later

1 Find and Select Namespaces

2 Assign Data Access Permissions for Selected Namespaces **1 Namespace Selected**

☒ Browse ☒ Read ☒ Write ☒ Delete ☐ Purge ☐ Privileged ☒ Search

☐ Read ACL ☐ Write ACL ☐ Change Owner

9. WORKING WITH BUCKET

This section demonstrates some basic operations on bucket. You can also perform these basic operations using S3 API call and tools (e.g. s3curl.pl, S3 browser) compatible with AWS S3.

9.1 ACCESS TO BUCKET

You can access the bucket through HCP Namespace Browser. The URL is in the form of:

Testing : “https://{bucket_name}.{gcis_id}.tlcosg-obj001.gcisdctr.hksarg”

Production: “https://{bucket_name}.{gcis_id}.obj001.gcis.hksarg”

1. Enter the URL of the bucket in the URL of the browser. A login page will be displayed. Enter username and password to log on.

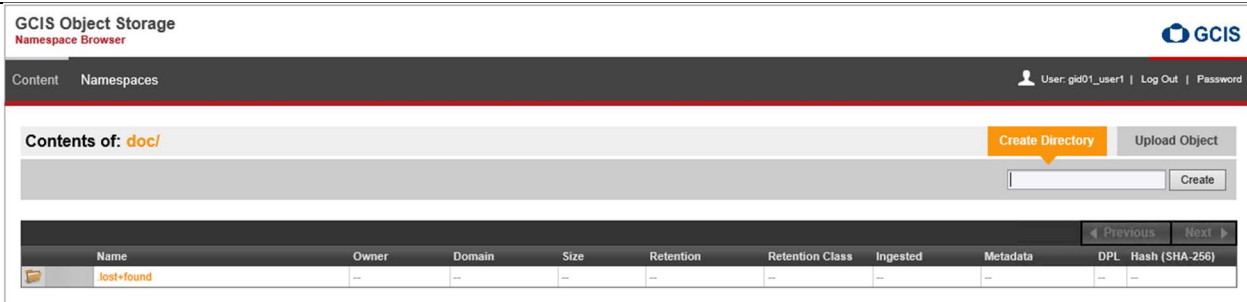
GCIS Object Storage
Namespace Browser

Username:

Password:

[License Information](#)

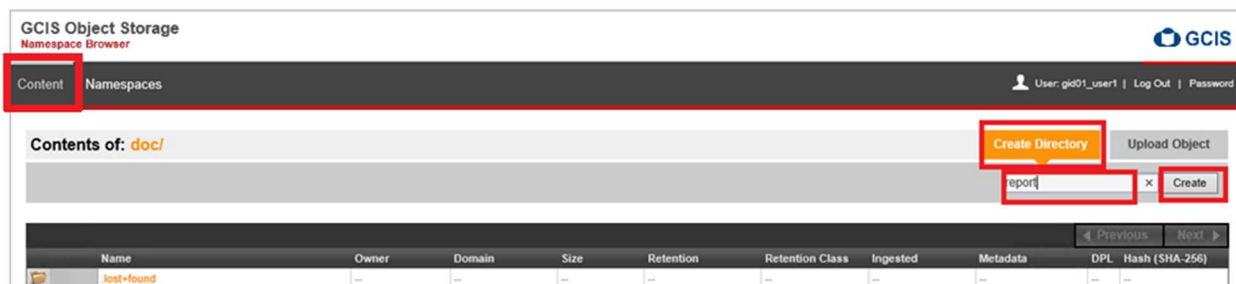
2. After successful log on, the bucket (namespace) main page is shown as follows.



9.2 CREATE DIRECTORY

For better management of your data, you can create directory to store specific category of data.

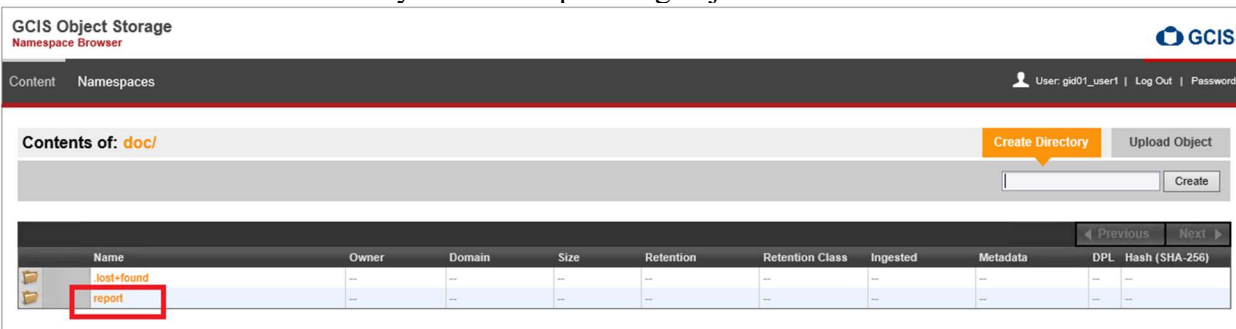
1. In the HCP Namespace Browser, click “Content”. Click “Create Directory”. Enter the directory name and then click “Create” button.



9.3 UPLOAD OBJECT

With “Write” permission, you can upload an object to a directory.

1. Click on the desire directory name for uploading object.



2. Click “Upload Object”. Click “Browse...” to choose file on your system. When you finish, click “Upload” to upload the object.

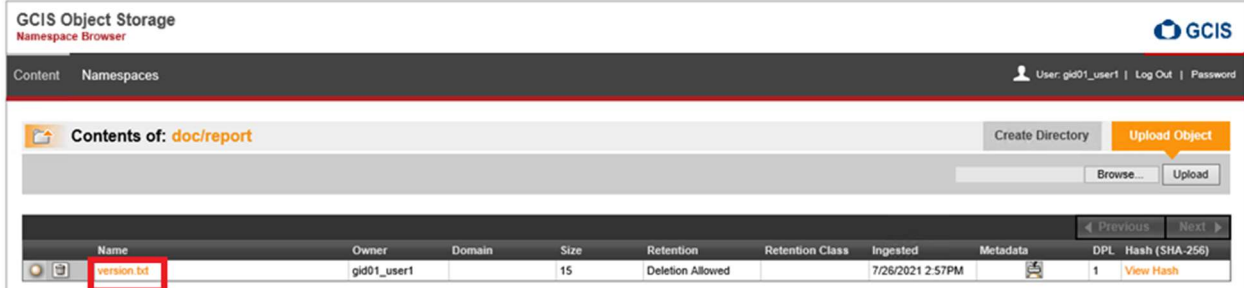


3. The object will appear on the list of objects in the directory.



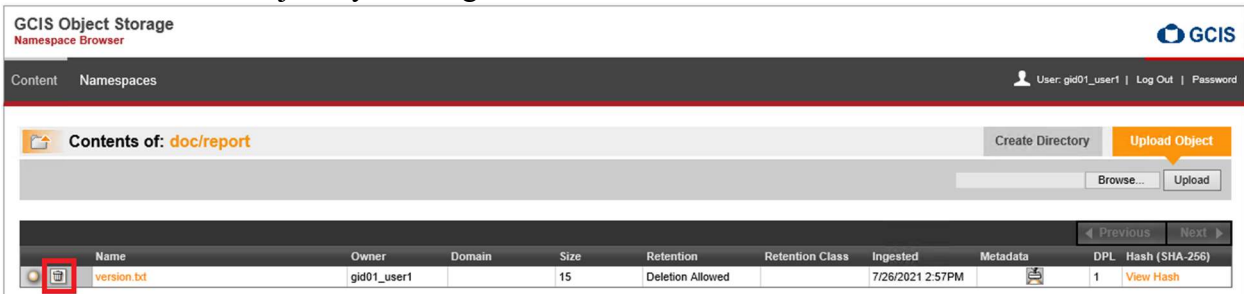
9.4 RETRIEVE OBJECT

Click on the object name to retrieve it. The browser will prompt you how to download the object.



9.5 DELETE OBJECT

You can delete the object by clicking on the trash icon.



10. DATA PROTECTION AND VERSIONING

10.1 WRITE ONCE, READ MANY (WORM)

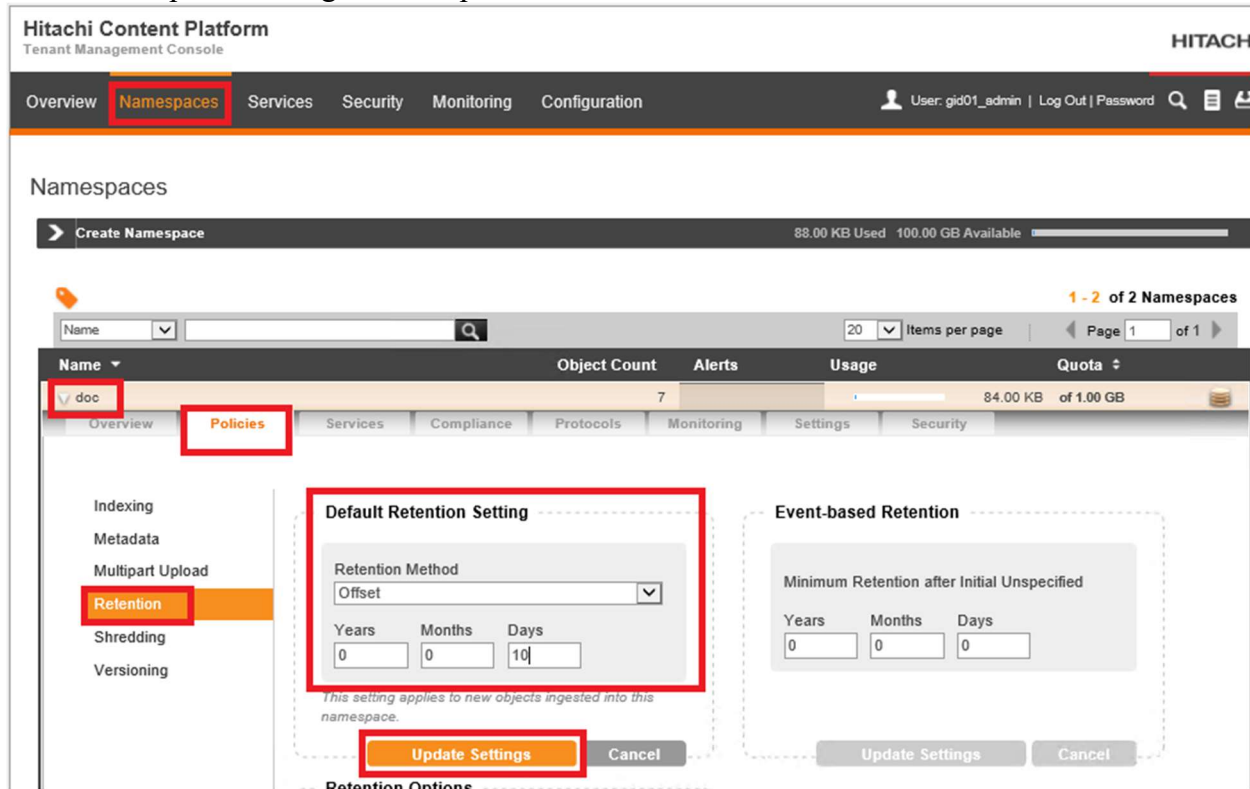
Tenants can create immutable data with WORM (Write Once, Read Many) feature to protect data for the retention period specified. During that period, the data can neither be rewritten nor deleted. This protection assures that the data will not be tampered with once it is written to the GCIS object storage.

Amazon S3 uses object lock to support WORM. However, object lock is not supported in GCIS object storage and tenant should use HCP WORM setting in bucket level configured through the Tenant Management Console so that all objects inside the bucket inherit the WORM configuration.

To enable WORM feature on a bucket:

1. Access HCP Tenant Management Console by using your administrative account and password.
2. The administrative account must have “Compliance” role in order to configure WORM feature on a bucket. Please refer to Section 7.1 “Add Administrator Role to a User” for adding “Compliance” role.
3. To configure WORM feature of a bucket, click “Namespace”. Choose the bucket from the namespace list. Click “Policies” tab. Click “Retention”

4. Enable the retention period by changing the “Default Retention Setting”:
 - a. Select “Offset” in “Retention Method”.
 - b. Enter the desired retention period in “Years”, “Months” and “Days”.
5. Click “Update Settings” to complete.



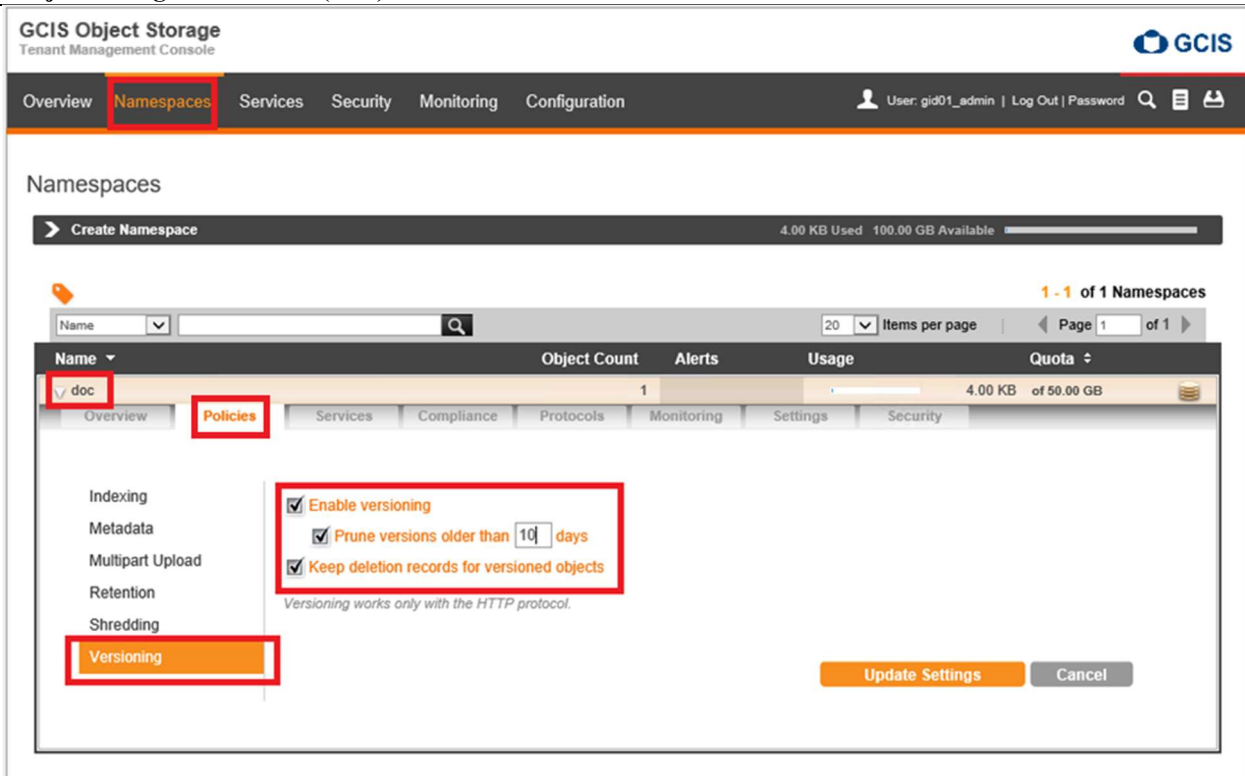
6. Now the data in the bucket cannot be deleted or replaced until after the retention period.

10.2 ENABLE VERSIONING ON A NAMESPACE (BUCKET)

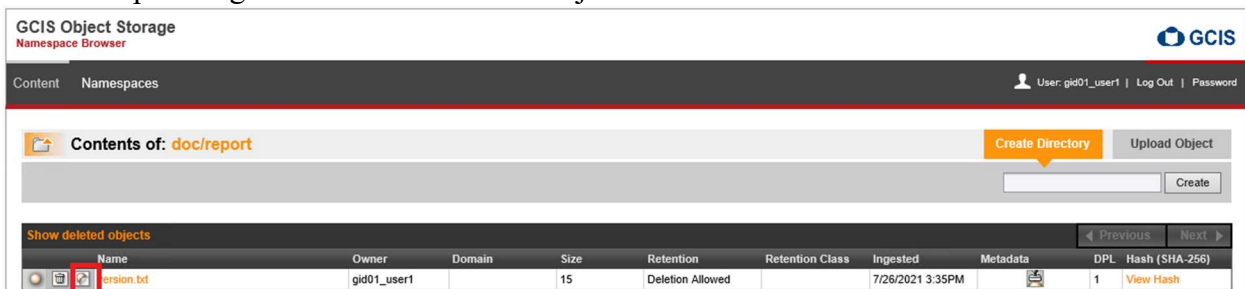
If the bucket you’re using is configured to allow versioning, you can store multiple versions of an object. You can use this technique, for example, to preserve multiple states of a file that changes over time.

Please note that HCP supports versioning on bucket-level. Objects inherit bucket’s versioning setting.

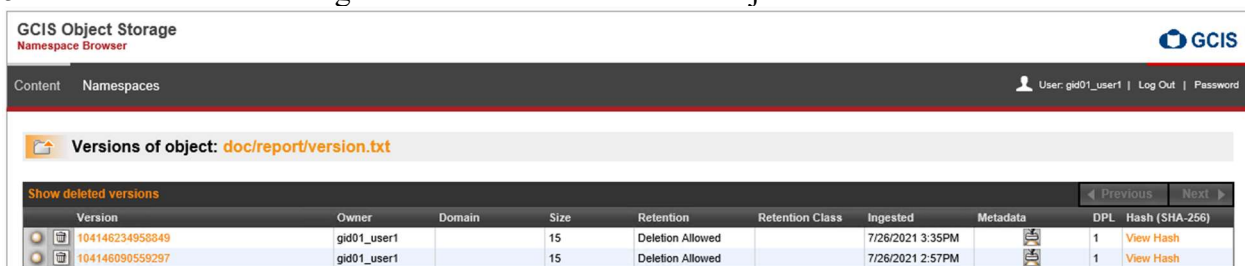
1. Access HCP Tenant Management Console by using your administrative account and password.
2. Click “Namespace”. Choose the bucket from the namespace list. Click “Policies” tab. Click “Versioning”.
 - a. Check on “Enable versioning”.
 - b. Check on and enter number of days in “Prune versions older than XX days” fields. This setting is the amount of time HCP should wait after a version is created before automatically deleting it. HCP automatically deletes only old versions of objects and never automatically deletes the most recent version. Please note that the housekeeping job runs at 0:00 every day.
 - c. If “Prune versions older than XX days” is not enabled, all object versions, including deleted versions, will remain in the bucket until you enable pruning or the objects are purged.
 - d. Check on “Keep deletion records for versioned object” for audit purpose.
3. Click “Update Settings” to complete.



4. Now you can upload new version of the existing object to the bucket. A versioning icon is shown after uploading the new version of the object.



5. Click on the versioning icon. The version list of the object is shown.



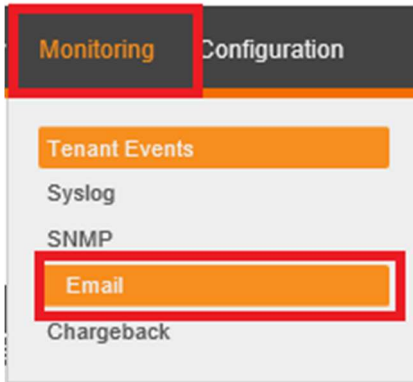
6. You can retrieve the specific version of the object by clicking on the version.

11. SETUP MONITORING ON BUCKET USAGE

Tenant can setup monitoring on bucket usage. When the usage of a bucket reaches the user-defined threshold, a notification email will be sent out.

11.1 CONFIGURE EMAIL NOTIFICATION

1. Log on to HCP Management Console. On the “Monitoring” menu item of the top menu bar, select “Email” to display the email page.



2. On the email page, select “Enable email notification”
 3. Optionally, change the format of the email to be sent in “Message Settings”. The content of the email messages HCP sends is determined by the message template specified in the Message Settings section on the Email page. You can modify this template at any time. The Message Preview section shows a sample email that uses the current template. In the From, Subject, and Body fields in the Message Settings section, specify the values that you want to use.
 4. The values you specify in the From, Subject, and Body fields in the email template can include variables that correspond to the information available for each log message (for example, the severity of the event that triggered the message or the short description of the event). When sending email, HCP replaces the variables in the email message with the applicable information.
 5. To include a variable in the email template, you specify the variable name preceded by the dollar sign (\$). A dollar sign followed by anything other than a variable name is displayed as a dollar sign in the email HCP sends. Please refer to Appendix “Email Template Variables” for the list of variables you can use in the email template.
 6. Specify one or more recipients to receive email about log messages in “Recipient”
 7. Each row in the Recipients section specifies:
 - a. One or more email addresses.
 - b. Whether to send email only about important log messages (Major) to the specified email addresses or to send email about all log messages (All).
 - c. The severity of the log messages about which to send email:
 - i. Notice tells HCP to send email about log messages with a severity level of notice, warning, or error.
 - ii. Warning tells HCP to send email about log messages with a severity level of warning or error.
 - iii. Error tells HCP to send email only about log messages with a severity level of error.
 - d. Whether to send email about general log messages (). General log messages are all messages that do not have a type of security or compliance.
 - e. Whether to send email about log messages with a type of security (.
 - f. Whether to send email about log messages with a type of compliance (.
- Email recipients receive email only about log messages that have all the selected properties.
7. Optionally, you send a test email by clicking on “Test”.
 8. Click “Update Setting” to complete.

GCIS Object Storage
Tenant Management Console

Overview Namespaces Services Security **Monitoring** Configuration

User: gid01_admin | Log Out | Password

Email

☒ **Enable email notification**

Note: by selecting this option, events will be sent through the SMTP server configured by the system administrator.

Message Settings

From: log@gid01.t1cosg-obj001.gciddctr.hksarg

Subject: [Severity] \$shortText

Body: Reason: \$reason

Reset

Message Preview

From: log@gid01.t1cosg-obj001.gciddctr.hksarg
 Subject: [Notice] Search enabled or disabled
 The following event occurred on Thu Aug 12 2021 9:10:31 AM HKT:
 Search enabled = true. Tenant: gid01

Reason:
A user enabled or disabled search.

Action:

Preview

Recipients mayyan@foo.com Add

Address	Importance	Severity			
dmchan@foo.com	Major	Error			

Test Update Settings Cancel

11.2 CONFIGURE BUCKET USAGE THRESHOLD

You can configure the bucket usage threshold for sending notification email.

1. Log on to HCP Tenant Management Console. Click “Namespace” on the top menu bar.
2. Click on “Modify namespace quota” icon of the bucket you want to change the usage threshold to display “Modify Namespace Quota” page.

GCIS Object Storage
Tenant Management Console

Overview **Namespaces** Services Security Monitoring Configuration

User: gid01_admin | Log Out | Password

Successfully deleted namespace. Dismiss

Namespaces

Create Namespace 92.00 KB Used 100.00 GB Available

1 - 2 of 2 Namespaces

Name	Object Count	Alerts	Usage	Quota
doc	8		88.00 KB of 1.00 GB	
pic	1		4.00 KB of 1.00 GB	

Modify namespace quota

3. On the “Modify Namespace Quota” page, the threshold is calculated in percentage of the hard quota of the bucket. Enter the usage threshold value in “Soft Quota %” field. Click “Update Quota” to finish.

Modify Namespace Quota
Updating quota for namespace 'doc'

Hard Quota: 1.00

Soft Quota %: 85

GB (selected) / TB

Close / Update Quota

12. ACCESS OBJECT STORAGE USING AWS S3 API

You can access GCIS object storage by making use of AWS S3 API call.

12.1 ENABLE ACLS AND HITACHI API

Before you can use AWS S3 API call to access GCIS object storage, you must enable ACLs setting, Hitachi API for Amazon S3 and management API settings.

1. Enable ACLs setting by clicking “Namespace”. Select the bucket (namespace). Click “Settings” tab, “ACLs”, check on the “Enable ACLs” and confirm with the popup.

GCIS Object Storage
Tenant Management Console

Overview **Namespaces** Services Security Monitoring Configuration

User: gid01_admin | Log Out | Password

Namespaces

Create Namespace 8.00 KB Used 100.00 GB Available

1 - 1 of 1 Namespaces

Name	Object Count	Alerts	Usage	Quota
doc	2		8.00 KB	of 50.00 GB

Overview Policies Services Compliance Protocols Monitoring **Settings** Security

ACLs

Compatibility

Retention Mode

Tags

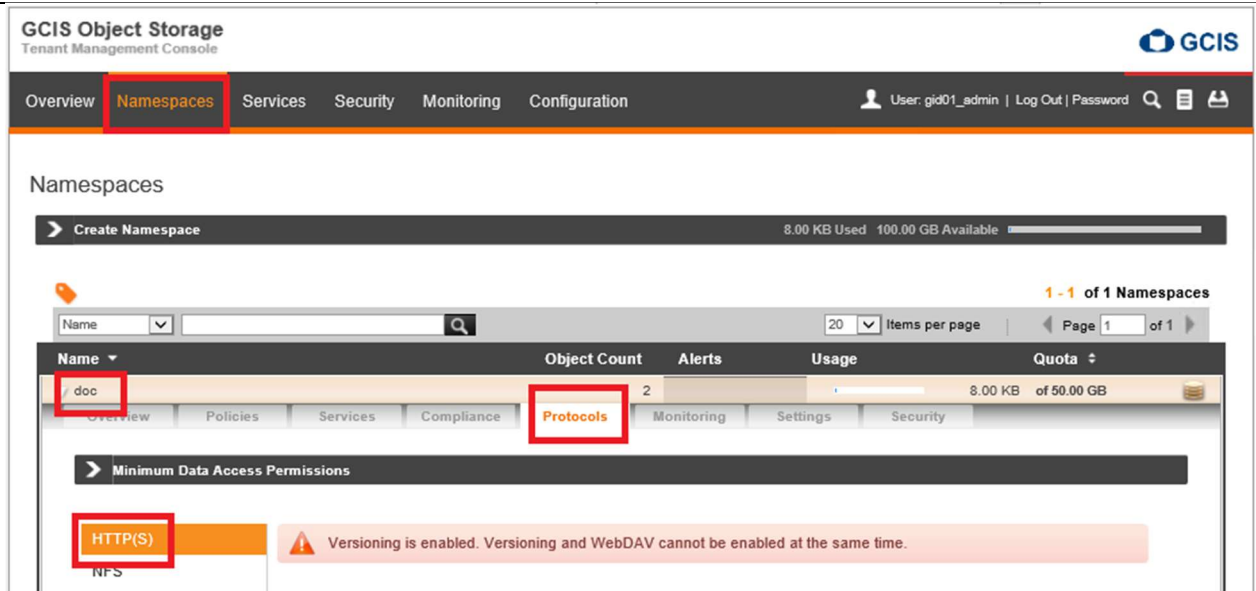
Name

Optimization

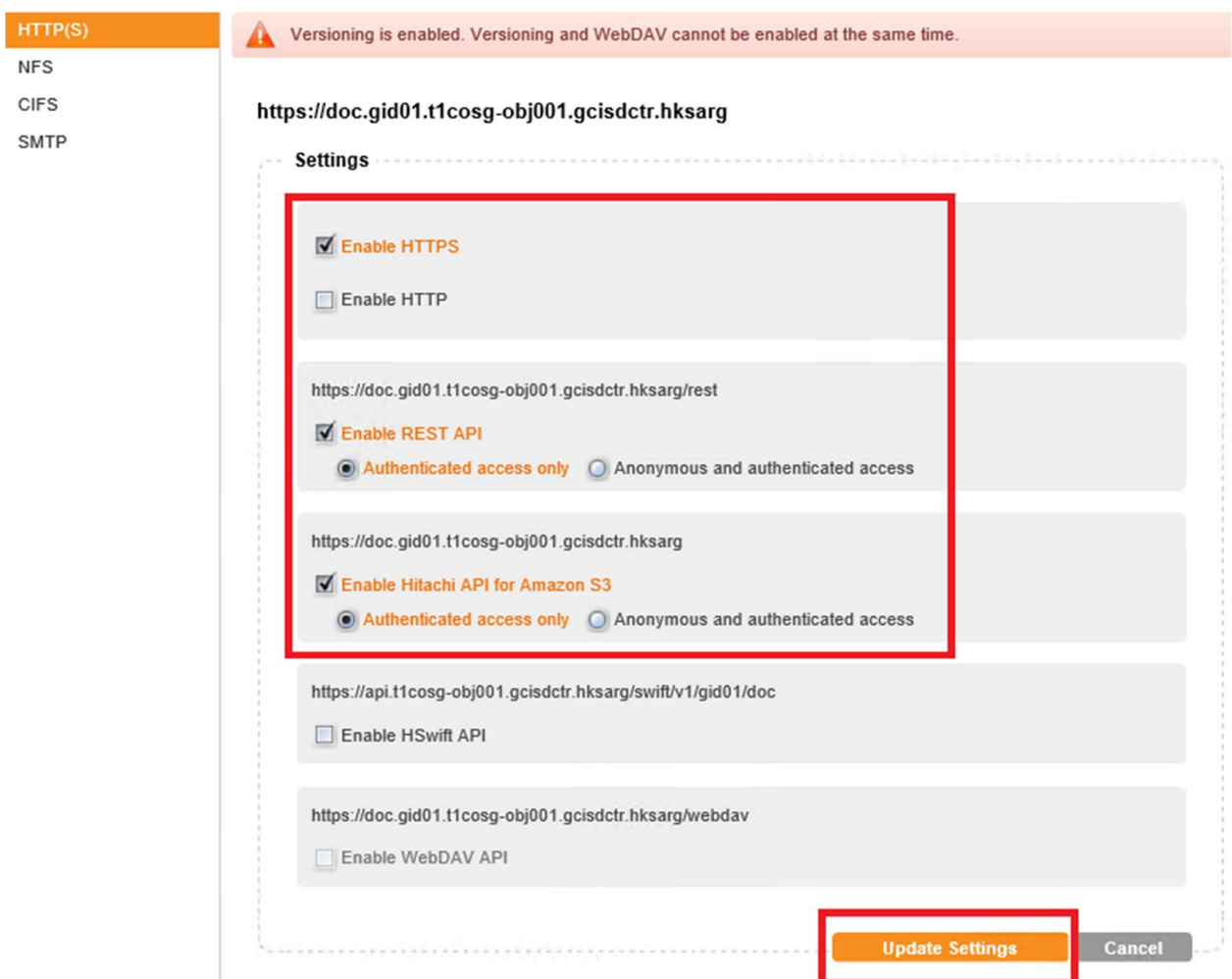
☒ Enable ACLs

Consistent ACLs cannot be disabled.

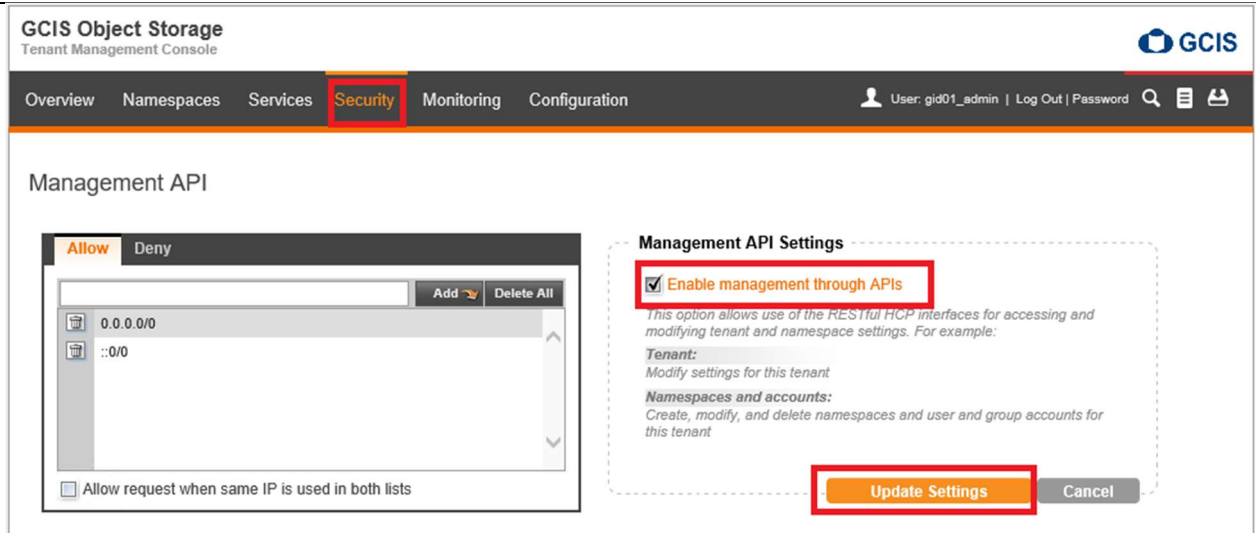
2. Next enable Hitachi API for Amazon S3 in protocols. Click “Namespace”. Choose the bucket (namespace) to configure. Click “HTTP(S)”.



3. Check on “Enable HTTPS”, “Enable REST API” and “Enable Hitachi API for Amazon S3”. Click “Update Settings” to finish.



4. Enable Management API Settings by clicking on “Security”. Check on “Enable management through APIs”. Click on “Update Settings” to finish.



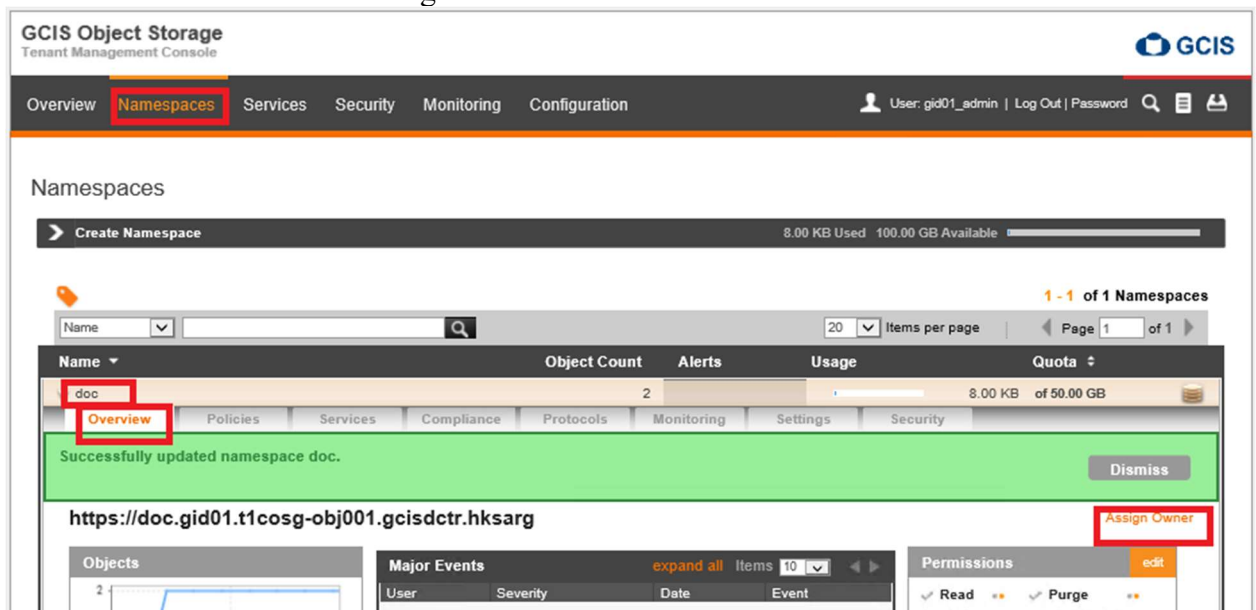
13. ACCESS OBJECT STORAGE USING S3 BROWSER

You can also access GCIS object storage using S3 browser.

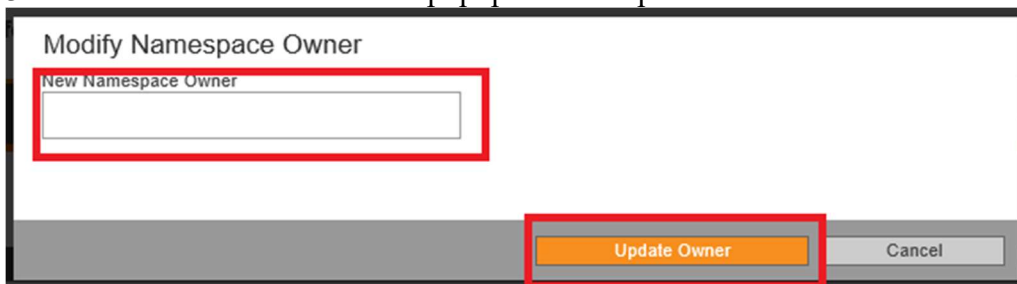
13.1 CHANGE BUCKET OWNER

S3 browser can only list buckets owned by the user. To change the owner of the bucket:

1. Log on to HCP Tenant Management Console with the admin user account.
2. Click "Namespace". Choose the namespace (bucket) you want to change the owner. Click "Overview" tab. Click "Assign Owner".



3. Enter the user account in the popup. Click "Update Owner" to finish.



13.2 ENABLE ACLS AND HITACHI API FOR AMAZON S3

Before you can use S3 browser to access GCIS object storage, you must enable ACLs setting, Hitachi API for Amazon S3 and management API Settings. Please follow the steps in Section 12.1.

13.3 RETRIEVE AUTHORIZATION TOKEN

Accessing GCIS object storage requires an authorization token as authorization to communicate with the HCP server.

An authentication token consists of a username in Base64-encoded format and a password that's hashed using the MD5 hash algorithm, separated by a colon, like this:

```
base64-encoded-username:md5-hashed-password
```

Example:

```
Z2lkMDFfYWRTaW4=:161ebd7d45089b3446ee4e0d86dbcf92
```

The GNU Core Utilities include the `base64` and `md5sum` commands necessary to encode your account information. These commands convert text to Base64-encoded and MD5-hashed values, respectively. For more information about the GNU Core Utilities, see <http://www.gnu.org/software/coreutils/>.

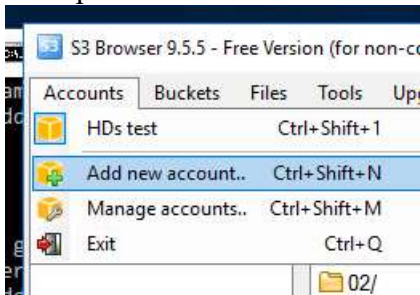
Other tools that generate Base64-encoded and MD5-hashed values are available for download on the web. For security reasons, do not use interactive public web-based tools to generate these values.

The GNU Core Utilities include the `base64` and `md5sum` commands, which convert text to Base64-encoded and MD5-hashed values, respectively. With these commands, a line such as this creates the required token:

```
echo `echo -n username | base64`:`echo -n password | md5sum` | awk  
'{print $1}'
```

13.4 SETUP IN S3 BROWSER

1. Open S3 browser. Click “Accounts” then select “Add new account..”



2. Fill the form as follows:
 - a. Enter your preferred account name in “Account Name”.
 - b. Select “S3 Compatible Storage” in “Account Type”.
 - c. Enter the your HCP Tenant Management Console namespace without the port number in “REST Endpoint”
 - d. Enter your base64-encoded user name in “Access Key ID”
 - e. Enter your md5 hashed password in “Secret Access Key”.
 - f. Click “Add new account to complete”

Add New Account [online help](#)

Enter new account details and click Add new account

Account Name:

Assign any name to your account.

Account Type:

Choose the storage you want to work with. Default is Amazon S3 Storage.

REST Endpoint:

Specify S3-compatible API endpoint. It can be found in storage documentation. Example: restserver.com:8080

Access Key ID:

Required to sign the requests you send to Amazon S3, see more details at <https://s3browser.com/keys>

Secret Access Key:

Required to sign the requests you send to Amazon S3, see more details at <https://s3browser.com/keys>

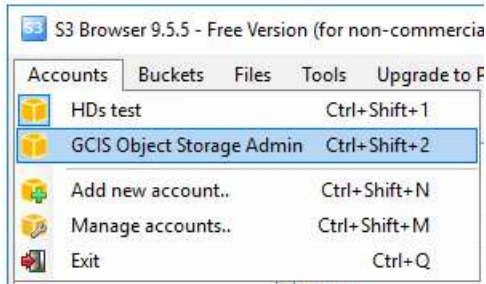
☐ Encrypt Access Keys with a password:

Turn this option on if you want to protect your Access Keys with a master password.

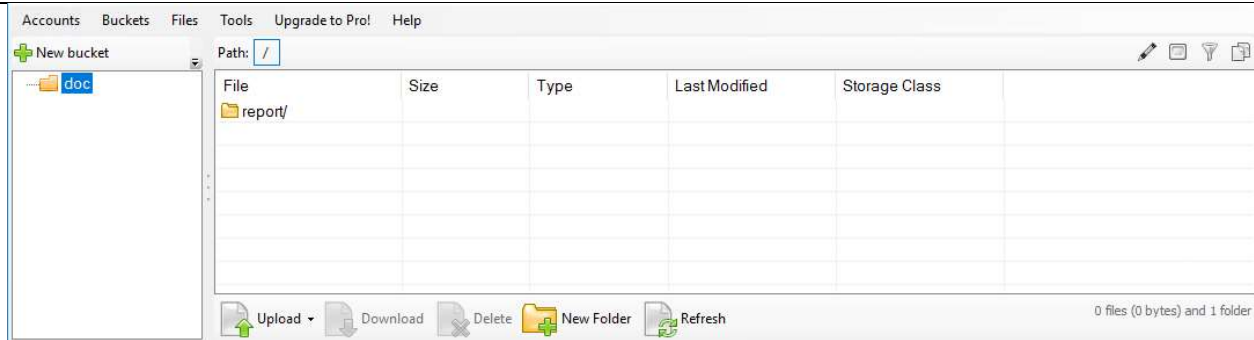
☒ Use secure transfer (SSL/TLS)
If checked, all communications with the storage will go through encrypted SSL/TLS channel

[Advanced S3-compatible storage settings](#)

3. Click “Accounts” in the S3 browser again. The new account is shown in the list.



4. Click on the new account. The S3 browser will connect to the GCIS object storage and it will retrieve the list of buckets owned by the user.



14. COMPATIBILITY OF AWS S3

The following http-based API is compatible with Amazon S3. B/Ds can use the third-party tools or library to call GCIS object storage.

Create buckets (PUT bucket)
** As there is no quota feature in Amazon S3 bucket, hard quota for the bucket is set to a default value that can be set through HCP Tenant Management Console when the bucket is created by S3 API. In order to set the hard quota when creating the bucket using API call, HCP management API should be used instead.
List the buckets you own (GET service)
Check the existence of a bucket (HEAD bucket)
Add ACLs to existing buckets (PUT bucket acl)
Retrieve ACLs for buckets (GET bucket acl)
Enable or suspend object versioning for buckets you own (PUT bucket versioning)
** If versioning is enabled for a bucket by S3 API, pruning is disabled by default. To set the pruning period when versioning is enabled, you can either use HCP management API or HCP Tenant Management Console to do this.
Check the status of object versioning for buckets you own (GET bucket versioning)
List objects that are in a bucket (GET bucket)
List versions of objects that are in a bucket (GET bucket versions)
Delete buckets you own, as long as the buckets don't have any objects in them (DELETE bucket)
Store objects in a bucket (PUT object)
Create folders in a bucket (PUT object/)
Add custom metadata to objects when uploading, where the custom metadata is specified as property/value pairs (PUT object copy replace)
** After you upload the object, you cannot modify object metadata. The only way to modify object metadata is to make a copy of the object and set the metadata.
Check the existence of an object or folder (HEAD object)
Retrieve custom metadata for objects (HEAD object)
Add ACLs to existing objects (PUT object acl)
Retrieve ACLs for objects (GET object acl)
Copy objects (PUT object copy)
Retrieve objects (GET object)
Delete objects (DELETE object)

15. ADD USER-DEFINED METADATA WHEN UPLOADING OBJECT

Assigning metadata to an object can only be done by uploading the object using S3 API call. The user-defined metadata in form of a name-value (key-value) pair. The user-defined metadata names must begin with "x-amz-meta-" to distinguish them from other HTTP headers.

The following is an example to upload object with user-defined metadata:

```
curl -X PUT -T "Q3_report.txt" -k \
-H "Authorization: HCP Z2lkMDFFyYWRtaW4=:161ebd7d45089b3446ee4e0d86dbcf92" \
-H "x-amz-meta-class: unclassified" \
"https://gid01.tlcosg-obj001.gcisdctr.hksarg/hs3/doc/report/Q3_report.txt"
```

16. LIMITATIONS AND CONSTRAINTS

1. GCIS object storage cannot be exported as a NFS or Samba for the sake of security.
2. A tenant can only create maximum of 20 buckets.
3. The quota of a bucket created through S3 API call or S3 Browser will be set by default value. To change the corresponding default quota of bucket, place mouse pointer over “Configuration” menu in HCP Tenant Management Console. Select “Namespace Defaults”. Change the default quota in “Hard Quota”. Click “Update Settings” to finish.

The screenshot shows the GCIS Object Storage Tenant Management Console interface. The top navigation bar includes 'Overview', 'Namespaces', 'Services', 'Security', 'Monitoring', and 'Configuration'. The 'Configuration' menu is highlighted. Below the navigation bar, the 'Namespace Defaults' section is visible. It includes a 'Description' field, a 'Hash Algorithm' dropdown set to 'SHA-256', a 'Hard Quota' section with a value of '50.00' and radio buttons for 'GB' and 'TB' (both are selected), and a 'Soft Quota' section with a value of '85' and a '%' radio button. Below these, there are four feature configuration sections: 'Cloud Optimized' (Off, On, System default), 'Retention Mode' (Enterprise, Compliance), 'Search' (Off, On), and 'Versioning' (Off, On). At the bottom right, there are 'Update Settings' and 'Cancel' buttons.

4. The ACL of an object cannot be set in the HCP Namespace Browser. The ACL can be configured using API call and third-party tools compatible with AWS S3.

17. APPENDIX

17.1 AVAILABLE ROLES

The roles that you can associate with a user or group account are:

Role	Description
Monitor	Grants permission to use the System Management Console to view the HCP system status and most aspects of the system configuration, including tenant configurations. The monitor

	role also grants read-only permission to view user and group accounts.
Administrator	Grants permission to use the System Management Console to view the HCP system status, perform most system configuration activities, create and manage tenants, and download the HCP internal logs. The administrator role also grants read-only permission to view user and group accounts. The administrator role does not grant permission to configure user or group accounts.
Security	Grants permission to use the System Management Console to view the HCP system status, create and manage user accounts, configure remote authentication, modify system security settings, configure syslog and SNMP logging and email notification, and view security events in the system log.
Compliance	Grants permission to use the Tenant Management Console to work with retention classes and retention-related settings and perform privileged deletes, as well as to use the System Management Console to view the HCP system status. Using the Tenant Management Console is possible only for the default tenant and for HCP tenants that are configured to allow system-level users to manage them and search their namespaces

For permissions granted by each role, please refer to

https://knowledge.hitachivantara.com/Documents/Storage/Content_Platform/8.3.x/Managing_a_Tenant_and_Its_Namespaces/Managing_accounts/01_About_user_and_group_accounts.

17.2 DATA ACCESS PERMISSIONS

Data access permissions allow users to access namespace content and some information about namespaces. These permissions are granted separately for individual namespaces.

The data access permissions that can be associated with user and group accounts for any given namespace are:

Permission	Description
Browse	List directory contents.
Read	- View and retrieve objects, including the system and custom metadata for objects. - View and retrieve previous versions of objects. - Check the existence of objects. - List annotations for objects. - For this permission to be granted, users must also have browse permission.
Read ACL	View and retrieve object ACLs.
Write	- Add objects to the namespace. - Modify system metadata (except retention hold). - Add or replace custom metadata.
Write ACL	Add, replace, and delete object ACLs.
Change owner	Change the owners of objects in the namespace.
Delete	Delete objects, custom metadata, and ACLs from the namespace.
Purge	Delete all versions of an object with a single operation. For this permission to be granted, users must also have delete permission.
Privileged	- Delete or purge objects that are under retention, provided the user also has delete or purge permission for the applicable namespace - Hold or release objects, provided the user also has write permission for the applicable namespace
Search	Use the HCP metadata query API and the HCP Search Console to query or search the namespace. For this permission to be granted, users must also have read permission.

**Users with any data access permissions for a namespace can view information about that namespace.

17.3 EMAIL TEMPLATE VARIABLES

The table below lists the variables you can use in the email template.

Variable	Description
\$action	The action to take in response to the message

GCIS object storage User Guide (v1.2)

\$date	The date and time at which the event occurred (for example, Wed Feb 8 2012 3:15:57 PM EST)
\$fullText	The full text of the message
\$id	The message ID
\$location	The fully qualified name of the HCP system on which the event occurred (for example, hcp-ma.example.com)
\$origin	For user-initiated events, the IP address from which the event request was sent and the port through which HCP received the event request, separated by a colon (for example, 192.168.152.181:8000)
\$reason	The reason why HCP issued the message
\$scope	
\$severity	The severity of the event that triggered the message
\$shortText	A brief description of the event that triggered the message
\$type	The type of message (General, Security, or Compliance), preceded by Important and a comma if the message is important (for example, Important, Security)
\$user	The user ID and username of the event initiator (for example, 105ff38f-4770-4f98-b5b3-8371ab0af359 lgreen)